The Pinnacles: Statues Carved by Time (Science — Earth & Space: Earth's changing surface)

TEACHER COPY — NOT FOR STUDENTS

Worksheet

Part A

1. Nouns: *rock, wind, desert* · Verbs: *crumble, blow, carry* · Adjectives: *jagged, enormous, ancient*
2. *The wind blew the sand across the beach and into the dunes.* (capital *T*; *blowed* → *blew*; missing full stop). Discuss irregular past tense.

Part C

1. Any two: about 200 km north of Perth; near Cervantes; in Nambung National Park; Western Australia.
2. Limestone, made from tiny pieces of seashells (from sea creatures) broken by waves.
3. Order: waves broke shells (1) → wind blew sand into dunes (2) → rain dissolved lime and made limestone (3) → plants grew; roots into cracks (4) → rainwater weathered limestone along cracks (5) → wind blew loose sand away (6).
4. Weathering: rock is broken down or worn away where it is. Erosion: the broken pieces are carried away by wind, water or ice.
5. E, W, E, W
6. The pillars were already there, hidden under sand, and the wind "unwrapped" them slowly, revealing them like a sculpture being uncovered; also they look like statues.
7. Any two: it would damage/break the fragile pillars; if every visitor took a rock there would be nothing left; it's the Traditional Custodians' Country; it's the law/rules of the park; future visitors deserve to see it.
8. They are still changing: wind still uncovers/buries pillars and rain and wind still wear away the limestone.

Connections

9. Yes — 1.75 + 1.75 = 3.5 (or 2 × 1.75 = 3.5).
10. About **2 hours**; return trip about **4 hours** in total driving.
11. Panel 4 shows both (weathering of limestone + wind erosion of sand); panel 1 also shows weathering-like breaking of shells by waves (accept); panel 2 shows wind carrying (eroding) sand. Reward reasoning.
12. In the Perth metropolitan area: the **Whadjuk Noongar** people. Check locally for other areas.

Part D

13. Models: ¶2 *made from broken seashells* · ¶3 *rain turned sand into limestone* · ¶5 *still slowly changing today* · ¶6 *people must protect the Pinnacles*

14. ¶1 — supports the idea that the pillars are many and varied in size / it's a strange landscape.
15. Main nouns: a) *landscape* b) *shells* c) *sea* d) *town*
16. e.g. *a huge, jagged limestone rock*; *the tall golden sand dune*; *a fierce, salty coastal wind*
17. a. jagged b) towering c) ancient d) fragile e) gritty

Part E

18. a-2, b-4, c-1, d-5, e-7, f-6 (3 is a distractor)
19. e.g. *Erosion by waves gradually washed the sand off the beach.*

Part F — marking notes

Strong paragraph: topic sentence (e.g. *The Pinnacles were formed over thousands of years by the sea, rain and wind.*); correct sequence with time connectives; weathering vs erosion used accurately; underlined noun groups; precise adjectives; concluding sentence (e.g. *…and they are still changing today.*). Check it is in the student's own words.

Part G

20. *like statues being unwrapped*; *looks like the surface of another planet*. Own similes, e.g. *The pillars stood like soldiers guarding the desert.*
21. It leaves the reader with the most important message — respect and protect the place; it moves from *how it formed* to *how we should behave*. The writer wants visitors to follow the rules and value the land.

Fast-finisher bank

F1. e.g. *the small fishing town of Cervantes*; *strong coastal winds*; *a sea of golden sand*; *the marked paths*; *the lower layers of the dunes*; *one careless climber*. Check ordering.
F2. E, W, W, E
F3. a) Both shaped by weathering then erosion; both in WA; both made of rock that formed long ago; both popular natural sights. b) Granite vs limestone; Wave Rock is one big cliff vs thousands of pillars; Wave Rock shaped underground by moist soil vs Pinnacles uncovered by wind; Wave Rock is far older. c) **Weathering first** — the shape began underground; *later* the soil was eroded away (signal word *later*).
F4. e.g. *The towering granite rock cast a long shadow. / We saw a breathtaking view of the river gorge. / The fierce coastal wind whipped sand across the path.*
F5. Look for: heading; 3 rules + reasons with *because / so that*; respectful acknowledgement (e.g. *This is Yued Noongar Country. Please treat it with respect.*).
F6. Check sequence (shell → sand → dune → limestone → weathered → uncovered → still changing), noun groups, personification, first person.
F7. e.g. 1) Spread sand in a tray. 2) Push small stones into the sand. 3) Blow gently through a straw across the sand. 4) Observe. Prediction: loose sand blows away, but sand directly behind/under the stones is protected and left in small mounds or columns — modelling how harder parts remain while loose sand is eroded.

Homework

1. A natural rock arch / hole in the rock in Kalbarri National Park, about 600 km north of Perth, above the Murchison River.
2. Some sandstone layers are softer; they were weathered and eroded faster, while harder layers above stayed.
3. Weathering: wind and rain weathered the weaker layers. Erosion: loose grains eroded away / flood water carries sand and pebbles.
4. e.g. *River floods deepen the gorge.*
5. Whole phrases are noun groups.
6. e.g. *the muddy, winding river*; *a slow brown river*
7. Pinnacles: limestone; mainly wind (plus rain weathering); thousands of pillars / made from shells. Nature's Window: sandstone; wind, rain and river water; a natural arch framing the river.
8. Any genuine example.

Teacher notes

- **Common misconceptions:** weathering and erosion used as synonyms; thinking the pillars were *built up* rather than *uncovered*.
- **If the student finishes early:** F3 → F7 can stretch beyond an hour; or start a "landform fact file" in `materials/fact-sheets/`.
- **If the student struggles:** read ¶4 aloud together; act out weathering (breaking a biscuit) vs erosion (sweeping the crumbs away).
- **Sources / fact-check:**

- Nambung National Park: ~200 km north of Perth near Cervantes; Pinnacles up to ~3.5 m; formed from shell-derived lime-rich sand → limestone; uncovered by wind. Yued (Noongar) Traditional Custodians. Managed by the WA Department of Biodiversity, Conservation and Attractions (Parks and Wildlife Service). Age phrased as “many thousands of years” because dating estimates vary.
- Wave Rock (Hyden): ~15 m high, ~110 m long, granite ~2.7 billion years old; flared slope from subsurface weathering then soil erosion.
- Kalbarri / Nature’s Window: Tumblagooda Sandstone (deposited over 400 million years ago); Murchison River gorge; Kalbarri ~570–590 km north of Perth by road (“about six hundred kilometres”).

- **Next week builds on:** topic sentences and noun groups → modal verbs and fact vs opinion in *Who Makes the Rules?* (civics — links to park rules in ¶6).